

Chapter 6

Covariant derivatives, Christoffel symbols & the Curvature tensor

6.1 Covariant derivatives

Under general coordinate transformations, the partial derivative of an $\binom{m}{n}$ tensor is not an $\binom{m}{n+1}$ tensor. We need to find a way of defining a covariant version of differentiation to keep our equations tensorial. We demand the following properties from our covariant derivative:

1. **Tensorial:** $D_\mu T^{\mu_1 \dots \mu_m}_{\nu_1 \dots \nu_n}$ is a component of an $\binom{m}{n+1}$ tensor for any $\binom{m}{n}$ tensor T .
2. **Reduce to partial derivatives for a scalar:** $D_\mu \phi = \partial_\mu \phi$ for any scalar field ϕ .
3. **Linearity:** $D_\mu(T + W) = D_\mu T + D_\mu W$ for any tensors T and W of the same rank.
4. **Product rule:** $D_\mu(TW) = (D_\mu T)W + T(D_\mu W)$ for any tensors T and W of the same rank.

Starting with a vector and requiring linearity, we have

$$D_\mu v^\rho = \partial_\mu v^\rho + \Gamma^\rho_{\mu\nu} v^\nu, \quad (6.1)$$

for some values of $\Gamma^\rho_{\mu\nu}$. These symbols are referred to as Christoffel symbols (of the second kind).

For a $\binom{2}{0}$ -tensor, we have

$$D_\mu T^{\alpha\beta} = \partial_\mu T^{\alpha\beta} + \Gamma^\alpha_{\mu\nu} T^{\beta\nu} + \Gamma^\beta_{\mu\nu} T^{\alpha\nu}.$$

Combining the product rule and the fact that D_μ of a scalar should be the same as $\partial_\mu u$, we get that the covariant derivative of a covector (1-form or $\binom{0}{1}$ -tensor) is

$$D_\mu v_\nu = \partial_\mu v_\nu - \Gamma^\rho_{\mu\nu} v_\rho.$$

Thus, we get the following prescription for the covariant derivative of a generic $\binom{m}{n}$ -tensor T :

1. Take the partial derivative of each component of T .
2. For each superscript, contract that superscript with the Christoffel symbol and add to the above.
3. For each subscript, contract that subscript with the Christoffel symbol and subtract from the above.

Note that D_μ is not an operator. Rather, it is a statement that a particular combination of tensor components along with their partial derivatives of a tensor transform as a tensor. That particular combination depends on the contravariant and covariant rank of the tensor.

Demystifying the covariant derivatives

$\partial_\mu v^\rho$ relates to change in the ρ component of a vector v if you move a little bit along the μ direction. $D_\mu v^\rho$ denotes the ρ component of the change in the vector v if you move a little bit along the μ direction. Note the difference between these two. They are not identical in general. The latter, involving the covariant derivative, relates to change in the geometrical objects and is therefore naturally tensorial.

6.2 Christoffel symbols

6.2.1 Transformation properties

For this combination in the RHS of (6.1) to transform like a $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ -tensor, the Christoffel symbols must have the following transformation rule:

$$\Gamma'^\rho{}_{\mu\nu} = \left(\frac{\partial x'^\rho}{\partial x^\alpha} \right) \left(\frac{\partial x^\beta}{\partial x'^\mu} \right) \left(\frac{\partial x^\gamma}{\partial x'^\nu} \right) \Gamma^\alpha{}_{\beta\gamma} + \left(\frac{\partial^2 x^\sigma}{\partial x'^\mu \partial x'^\nu} \right) \left(\frac{\partial x'^\rho}{\partial x^\sigma} \right). \quad (6.2)$$

to guarantee that the appropriate tensorial transformation of $D_\mu v^\rho$. If the unprimed coordinates are such that they are local inertial frames (i.e., flat space) at the chosen point, then $\Gamma^\rho_{\mu\nu} = 0$ for all μ, ν, ρ . For other coordinates, one can compute the various partial derivatives and use (6.2) to find the Christoffel symbols.

Exercise: Using $\frac{\partial}{\partial x'^\mu} = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial}{\partial x^\alpha}$, show that (6.2) is a necessary condition for $D_\mu v^\rho$ to transform like a $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ -tensor, i.e.,

$$(D_\mu v^\rho)' = \left(\frac{\partial x'^\rho}{\partial x^\alpha} \right) \left(\frac{\partial x^\beta}{\partial x'^\mu} \right) D_\beta v^\alpha.$$

Note on the definition

Rather than defining the Christoffel symbols as $\Gamma^\rho_{\mu\nu} = \text{blah}$, which would look very comforting, we are using some what of a recursive definition based on its expected transformation properties. Equation (6.2) is in a way a recursive relation with a reference value of 0 in Cartesian coordinates defined on a (locally) flat space. This leaves the definition of Christoffel symbols somewhat vague, but it is very much intentional. Some textbooks define the Christoffel symbols through the metric

and its first derivatives. I defer that for as long as possible (until Section 6.2.3) for two reasons - one practical and one philosophical:

1. In practice, we almost never use that relation to compute Christoffel symbols.
2. Such a definition gives the impression that the Christoffel symbols are a function of the metric, when in fact, Christoffel symbols can be defined even in spaces when the metric is not defined.

Its transformation properties defined above are more fundamental and primitive. When a metric is given, consistency requires that the Christoffel symbols are related in a specific manner to the metric and its first derivatives. Treating that as a definition is not a very justified stance. In GR, we deal with a manifold with a metric defined, so it is not incorrect to define it.

The Christoffel symbols of the first kind are

$$\Gamma_{\rho\mu\nu} = g_{\rho\alpha}\Gamma^\alpha_{\mu\nu}.$$

6.2.2 Comment on the symmetry of Christoffel symbols

If this is the first time you are reading about Christoffel symbols, feel free to skip this subsection.

6.2.3 Linking Christoffel symbols to the metric

To link the covariant derivative to the metric (and its partial derivatives), we can start with the fact that the covariant derivative of the metric is zero.

$$D_\rho g_{\mu\nu} = \partial_\rho g_{\mu\nu} - \Gamma^\alpha_{\rho\mu} g_{\alpha\nu} - \Gamma^\alpha_{\rho\nu} g_{\mu\alpha} = 0. \quad (6.3)$$

The above equation has two different Christoffel symbols appearing. To isolate just one of them we can write $\Gamma^\alpha_{\mu\nu}$ = something, we need to get a few more of these relations. Repeating the indices cyclically, we get

$$D_\mu g_{\nu\rho} = \partial_\mu g_{\nu\rho} - \Gamma^\alpha_{\mu\nu} g_{\alpha\rho} - \Gamma^\alpha_{\mu\rho} g_{\nu\alpha} = 0. \quad (6.4)$$

and

$$D_\nu g_{\rho\mu} = \partial_\nu g_{\rho\mu} - \Gamma^\alpha_{\nu\rho} g_{\alpha\mu} - \Gamma^\alpha_{\nu\mu} g_{\rho\alpha} = 0. \quad (6.5)$$

Subtracting the last two equations from the first of these, moving the terms without the Christoffel symbols to the right and using the symmetry of the metric, we get

$$g_{\rho\alpha} (\Gamma^\alpha_{\mu\nu} + \Gamma^\alpha_{\nu\mu}) + g_{\alpha\mu} (\Gamma^\alpha_{\rho\nu} - \Gamma^\alpha_{\nu\rho}) + g_{\alpha\nu} (\Gamma^\alpha_{\rho\mu} - \Gamma^\alpha_{\mu\rho}) = -\partial_\rho g_{\mu\nu} + \partial_\nu g_{\rho\mu} + \partial_\mu g_{\rho\nu} \quad (6.6)$$

There exist a number of possible Γ s that satisfies this relation. For a given ρ, μ, ν , this is one equation with more than one unknowns (Γ s) here. Even if we considered other values of ρ, μ, ν , in n -dimensions, we only have $n^2(n+1)/2$ equations of the kind $D_\rho g_{\mu\nu} = 0$ but n^3 unknowns ($\Gamma^\rho_{\mu\nu}$ s). This is an underconstrained system of linear equations, with infinitely many solutions. To find a unique solution, we need to introduce additional constraints.

A common choice is to require that the Christoffel symbols are symmetric under the exchange of its subscripts. This is known as the *torsion-free* condition (see Sec. 6.2.2 for some nuanced discussion).

If we impose this condition, this reduces the number of unknowns to $n^2(n+1)/2$. This doesn't change the physics, since the Christoffel symbols usually come contracted with a symmetric object (e.g., in geodesic equation). Moreover, starting with $\Gamma^\alpha_{\mu\nu} = 0$, the transformation relation ((6.2)) shows that the Christoffel symbols are indeed symmetric under the exchange of its subscripts. So the torsion-free assumption is a very natural one.

With this additional condition, the first two terms on the LHS cancel and we get

$$2g_{\rho\beta}\Gamma^\beta_{\mu\nu} = \partial_\nu g_{\rho\mu} + \partial_\mu g_{\rho\nu} - \partial_\rho g_{\mu\nu}. \quad (6.7)$$

We replaced the dummy index α with β above. Contracting it with the inverse metric, $g^{\alpha\rho}$, we get

$$\Gamma^\alpha_{\mu\nu} = \frac{1}{2}g^{\alpha\rho}(\partial_\nu g_{\rho\mu} + \partial_\mu g_{\rho\nu} - \partial_\rho g_{\mu\nu}). \quad (6.8)$$

Thus, Christoffel symbols are related to the metric through the following relation:

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2}g^{\rho\alpha}(\partial_\mu g_{\alpha\nu} + \partial_\nu g_{\alpha\mu} - \partial_\alpha g_{\mu\nu}). \quad (6.9)$$

As a check for self-consistency, we note that the RHS of this equation is symmetric under the exchange of μ and ν .

Exercise: Given that the transformation property is the key in the definition of the Christoffel symbols, show that the RHS of (6.9) transforms as expected in (6.2).

6.3 Interpretation of covariant derivatives and Christoffel symbols in flat space

The discussion here is for a generic n -dimensional Euclidean space. As you read this, it would help to keep polar coordinates in $\mathbb{R}^2$ in the back of your mind as a concrete example.

Consider some vector field $\vec{v}(x^1, x^2, \dots, x^n)$. We could ask how this vector field changes as we move along the x^i direction. The rate of change is given by the vector

$$\lim_{\Delta x^i \rightarrow 0} \frac{\vec{v}(x^1, x^2, \dots, x^i + \Delta x^i, \dots, x^n) - \vec{v}(x^1, x^2, \dots, x^n)}{\Delta x^i}. \quad (6.10)$$

If these things converge well (which is why we needed this to be topological space), then we can also write it as

$$\lim_{\Delta x^i \rightarrow 0} \frac{\vec{v}(x^1, x^2, \dots, x^i + \Delta x^i/2, \dots, x^n) - \vec{v}(x^1, x^2, \dots, x^i - \Delta x^i/2, \dots, x^n)}{\Delta x^i} := D_i \vec{v}. \quad (6.11)$$

For a given i , this is a vector defined at the point $(x^1, x^2, \dots, x^n)$.

$$(D_i \vec{v}) = \partial_i (v^j(x) \hat{e}_j(x)) = [\partial_i v^j(x)] \hat{e}_j(x) + v^j(x) [\partial_i \hat{e}_j(x)]. \quad (6.12)$$

If we express the vector $\partial_i \hat{e}_j(x)$ as linear combinations of basis vectors at that point

$$\frac{\partial \hat{e}_i}{\partial x^j} = \Gamma^l_{ij}(x) \hat{e}_l, \quad (6.13)$$

where $\Gamma^k_{ij}(x)$ are the Christoffel symbols at the point $(x^1, x^2, \dots, x^n)$.

If we ask for a particular k^{th} component of $D_i \vec{v}$, we get

$$(D_i \vec{v})^k = \partial_i v^k(x) + v^j(x) \Gamma^k_{ij}(x). \quad (6.14)$$

Unfortunately, we denote this k^{th} component as $D_i \vec{v}$ as $D_i v^k$ instead of $(D_i v)^k$ which makes it look like we are doing something to one component of the vector, when we really differentiating the vector as a whole along a particular coordinate. This is one place where thinking of geometrical objects instead of the components turns out to be useful.

Lets denote the points $(x^1, x^2, \dots, x^i + \Delta x^i/2, \dots, x^n)$ as P and $(x^1, x^2, \dots, x^i - \Delta x^i/2, \dots, x^n)$ as Q . What if we considered In n -dimensions, instead of computing n^3 (at least $\mathcal{O}(n^3)$) Christoffel symbols, we can compute n^2 derivatives of the basis vectors and read off the Christoffel symbols. Later, we will see how to read them off from just n equations (c.f. Eq. (7.1))!

This argument is valid only in flat space, which we haven't defined yet. In curved spaces, the Christoffel symbols still capture the change to the basis vectors, but not because we chose non-Cartesian coordinates but because the vector space in which all the vectors (including the basis vectors) lie itself changes. This is certainly a subtlety but this interpretation generalizes well. This will need to wait on the discussion of manifolds and tangent planes.

But more importantly, we ask how does $D_i \vec{v}$ look in another coordinate system (primed). In general, the separation between P and Q will not be along any particular coordinate direction.

$$D_i \vec{v} = \lim_{d(P,Q) \rightarrow 0} \frac{1}{\Delta x^i} (\vec{v}(x'^1 + \Delta x'^1/2, \dots, x'^n + \Delta x'^n/2) - \vec{v}(x'^1 - \Delta x'^1/2, \dots, x'^n - \Delta x'^n/2)) \quad (6.15)$$

$$= \lim_{d(P,Q) \rightarrow 0} \frac{1}{\Delta x^i} (\Delta x'^a D'_a(\vec{v})) \quad (6.16)$$

$$= \frac{\partial x'^a}{\partial x^i} D'_a(\vec{v}) \quad (6.17)$$

The last line shows that D_i itself transforms like a $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ tensor. If we further work out how a specific k^{th} component of $D_i \vec{v}$ look like in terms of the components in the primed coordinates, we will see that $D_i v^k$ transforms like a $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ tensor.

Exercise: Obtain the Christoffel symbols for polar coordinates in $\mathbb{R}^2$ by considering how the basis vectors $\hat{e}_\rho$ and $\hat{e}_\varphi$ change.

6.4 Curvature tensor

The covariant derivatives seem to have almost all the nice properties of partial derivatives, but better because they take in a tensor and give another tensor. This seems too good to be true. What are we missing?

The price we have paid is that the covariant derivatives don't commute with each other necessarily. For any tensor T , we have

$$\partial_\mu \partial_\nu T = \partial_\nu \partial_\mu T.$$

Rewriting it differently,

$$[\partial_\mu, \partial_\nu] T := \partial_\mu \partial_\nu T - \partial_\nu \partial_\mu T = 0,$$

where we have introduced the commutator $[A, B] := AB - BA$. Let us look at the commutator of the covariant derivatives. For concreteness, let's take a vector field v (we know it reduces to partial derivatives for scalars, so we won't learn anything there).

We are guaranteed that $D_\mu D_\nu v^\rho - D_\nu D_\mu v^\rho$ must be a $\binom{1}{2}$ -tensor. We are also guaranteed that that tensor is linear in the vector. So it must be of the form

$$[D_\mu, D_\nu] v^\rho := D_\mu (D_\nu v^\rho) - D_\nu (D_\mu v^\rho) = R^\rho{}_{\sigma\mu\nu} v^\sigma, \quad (6.18)$$

where $R^\rho{}_{\sigma\mu\nu}$ are the components of a $\binom{1}{3}$ indexed object R . We have done the hard work of developing the covariant derivative upfront, so now we can coast now and declare that R is a tensor. R is manifestly a tensor, since the other quantities in the equations are. This is the Riemann curvature tensor and plays a key role in GR.

We can express the components of the Riemann curvature tensor in terms of the Christoffel symbols as

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho{}_{\nu\sigma} - \partial_\nu \Gamma^\rho{}_{\mu\sigma} + \Gamma^\rho{}_{\mu\lambda} \Gamma^\lambda{}_{\nu\sigma} - \Gamma^\rho{}_{\nu\lambda} \Gamma^\lambda{}_{\mu\sigma}. \quad (6.19)$$

Exercise: Express the covariant derivatives in terms of the Christoffel symbols and derive this expression for the components of the Riemann curvature tensor.

Exercise: Using the transformation properties of the Christoffel symbols (Eq. 6.2), show that the Riemann curvature tensor transforms as a $\binom{1}{3}$ -tensor, i.e.,

$$R'^\rho{}_{\sigma\mu\nu} = \frac{\partial x'^\rho}{\partial x^\alpha} \frac{\partial x^\beta}{\partial x'^\sigma} \frac{\partial x^\gamma}{\partial x'^\mu} \frac{\partial x^\delta}{\partial x'^\nu} R^\alpha{}_{\beta\gamma\delta}. \quad (6.20)$$

6.4.1 Significance of the Riemann curvature tensor

Why is this called the *curvature* tensor? In flat space(time), it is possible to set a global Cartesian coordinate system where the Christoffel symbols vanish identically everywhere. Thus, the Riemann curvature tensor vanishes identically in flat space(time). By the *once zero, always zero* mantra, the components $R^\rho{}_{\sigma\mu\nu}$ are zero in flat space(time) in any coordinates. Equivalently, a vanishing

Riemann tensor everywhere indicates that the space is flat even if that's not apparent from the coordinates. Thus, we finally have a way of quantifying what we mean by a curved space. A departure from zero indicates the local curvature of the space.

Another way of seeing it is that, this tensor involves second derivatives of the metric. Second derivatives of a function indicate the curvature of that function (at the point where the second derivatives are evaluated). If we consider the metric to be denoting the space itself, then its second derivatives capture the curvature of space.

This is also interesting from the physics point of view. From our schematic description of the Einstein equation, we are looking for a tensorial object that involves the second derivatives of the metric. We had a mild hurdle when we realized the derivatives of a tensor don't yield a tensor. But we finally have a tensor that is made of second derivatives of the metric. We anticipated a rank-2 tensor, but what we currently have is a rank-4 tensor. We will obtain a rank-2 tensor from it soon, but let's analyze this tensor first.

Pnemonic

Equation 6.19 is admittedly daunting unless you've stared at it for long enough (and one semester is not long enough). A pnemonic to get the indices right is as follows:

1. R is a $\binom{1}{3}$ -tensor. Christoffel symbols are $\binom{1}{2}$ indexed objects (not tensors).
2. Schematically, $R \sim \partial\Gamma + \Gamma\Gamma$
3. $R^\rho{}_{\sigma\mu\nu}$ is antisymmetric under the exchange of the last two indices μ and ν .
4. The indices must be according to the Einstein summation convention.

There is just one way of writing it once you get these right.

6.4.2 Properties of the Riemann curvature tensor

Given its significance, it is important to know some of its properties. Having it as a $\binom{1}{3}$ tensor is a bit odd. It is often convenient to work with the Riemann tensor contracted with the metric or its inverse, so that either it is a fully covariant $\binom{0}{4}$ tensor with components $R_{\rho\sigma\mu\nu}$ or a mixed $\binom{2}{2}$ tensor with components $R^{\rho\sigma}{}_{\mu\nu}$.

1. $R_{\rho\sigma\mu\nu} = -R_{\rho\sigma\nu\mu}$. Equivalently, $R^{\rho\sigma}{}_{\mu\nu} = -R^{\rho\sigma}{}_{\nu\mu}$. The antisymmetry in μ and ν is evident from the antisymmetry of the commutator and is present regardless of whether the other indices are both superscripts or subscripts or mixed.
2. $R_{\rho\sigma\mu\nu} = -R_{\sigma\rho\mu\nu}$. Or equivalently, $R^{\rho\sigma}{}_{\mu\nu} = -R^{\sigma\rho}{}_{\mu\nu}$. This antisymmetry does not appear in the $\binom{1}{3}$ version.
3. $R^{\rho\sigma}{}_{\mu\nu} = R^{\mu\nu}{}_{\rho\sigma}$ or equivalently $R_{\rho\sigma\mu\nu} = R_{\mu\nu\rho\sigma}$. R is symmetric under the exchange of the first pair of indices with the second pair.
4. $R^\rho{}_{\mu\nu\sigma} + R^\rho{}_{\nu\sigma\mu} + R^\rho{}_{\sigma\mu\nu} = 0$.

Often, it is convenient to choose the locally inertial frame where the Christoffel symbols vanish (but not its derivatives) to obtain a simpler expression for R .

$$R^\rho{}_{\sigma\mu\nu}(x)\Big|_{\text{LIF}} = \partial_\mu \Gamma^\rho{}_{\nu\sigma}(x) - \partial_\nu \Gamma^\rho{}_{\mu\sigma}(x)\Big|_{\text{LIF}}. \quad (6.21)$$

Exercise: Show that the properties of the Riemann tensor listed above are true. You may work in the LIF. *Hint:* Substitute the expression for Christoffel symbols from Eq. (6.9) first, take the derivatives and then set the first derivatives of the metric to zero.

6.4.3 Number of independent components of the Riemann tensor

It is an interesting exercise in combinatorics to find the number of independent components of the Riemann tensor in n -dimensions. Let's call it $C(n)$. We'll specialize to $n = 4$ later. A generic tensor of rank- r in n dimensions has n^r components.

Let's start with the pairwise symmetry of the indices. If (ρ, σ) and (μ, ν) can take N values, then the number of independent components is $\frac{1}{2}N(N+1)$. The antisymmetry in (μ, ν) and (ρ, σ) implies that $N = n(n-1)/2$. This implies that from the first three conditions alone, the number of independent components is $\frac{1}{2}(n(n-1)/2)(n(n-1)/2 + 1) = \frac{1}{8}(n^4 - 2n^3 + 3n^2 - 2n)$. The fourth condition, along with the earlier ones, implies that $R_{\rho\sigma\mu\nu}$ is fully antisymmetric in all four of its indices. This implies that we have overcounted the number of components by $n(n-1)(n-2)(n-3)/4!$. Note that this overcounting is absent for $n < 4$. Subtracting this from the earlier count, we get

$$C(n) = \frac{1}{12}n^2(n^2 - 1) \quad (6.22)$$

Unsurprisingly, this is the same number of second derivatives that we could not set to zero by a convenient choice of coordinates. This is not a coincidence and is reflecting the fact that the Riemann tensor can only depend on those 20 numbers (for $n = 4$) in some manner. Thanks to these four properties, we have a significant reduction to 20 from 256 possible components.

n	$C(n)$	n^4
1	0	1
2	1	16
3	6	81
4	20	256

Table 6.1: Number of independent components of the Riemann tensor in n dimensions

There is no curvature in 1-dimension. In 2-dimensions, a single number captures the curvature.

Exercise: Consider the metric on the sphere $\mathbb{S}^2$ given by the line element

$$dl^2 = a^2 (d\theta^2 + \sin^2 \theta d\phi^2), \quad (6.23)$$

for some positive constant a . a has the interpretation of the radius of the sphere when $\mathbb{S}^2$ is embedded in $\mathbb{R}^3$. Obtain the single independent component of the Riemann curvature tensor.

Exercise: Consider the surface of a cylinder given by the line element

$$dl^2 = a^2 d\theta^2 + dz^2, \quad (6.24)$$

for some positive constant a . a has the interpretation of the radius of the cylinder when it is embedded in $\mathbb{R}^3$. Obtain the single independent component of the Riemann curvature tensor.